

Project Task Distribution and Timeline

Oscillating Water Column (OWC) Numerical Study

The project is divided into three main components, each assigned to one student. While each student has a primary responsibility, all members are expected to understand the overall workflow and support integration. The work is structured so that all three components can progress in parallel.

1. Hydrodynamics (Wave Tank and OWC Geometry)

This component focuses on setting up the numerical wave tank and representing the physical structure of the Oscillating Water Column (OWC). The objective is to ensure that the wave motion and its interaction with the chamber are physically realistic.

Weeks 1–2

- Start from the Basilisk example `simple_waveflume.c`.
- Define the computational domain (length, depth) and initialise the water level.
- Implement regular wave generation at the inlet boundary.
- Run the simulation without any structure and verify that waves propagate cleanly across the domain with minimal reflection.

Weeks 3–4

- Introduce the OWC chamber using an embedded boundary (solid geometry).
- Ensure that the free surface inside the chamber oscillates smoothly and responds to incoming waves.
- Check that the presence of the chamber does not introduce numerical instability.

Deliverable

- A working Basilisk simulation of the wave tank with OWC geometry.
- Plots showing free-surface elevation (wave profiles) at selected locations.

2. PTO Modelling (Power Extraction)

This component focuses on modelling how energy is extracted from the oscillating air column above the water inside the chamber. A simplified Power Take-Off (PTO) model will be used.

Weeks 2–4

- Extract pressure inside the OWC chamber as a function of time, $\Delta P(t)$.
- Define a simple PTO relationship between pressure and airflow, such as $Q = f(\Delta P)$.
- Compute instantaneous power using:

$$P(t) = \Delta P(t) \cdot Q(t)$$

- Ensure that the computed quantities are consistent and physically reasonable.

Weeks 5–6

- Couple the PTO model with the hydrodynamics simulation.
- Run simulations for different damping parameters and observe how the system responds.
- Identify expected trends (e.g. changes in amplitude and phase).

Deliverable

- A module or script that computes power output.
- Plots of time-averaged power $\bar{P}$ as a function of PTO damping.

3. Wave Analysis (Transmission and Reflection)

This component focuses on analysing how the OWC modifies the incoming wave field. When waves encounter a structure, part of the wave energy is reflected back, part is transmitted downstream, and part is absorbed by the system. The aim here is to quantify these effects using measurable wave properties.

Basic concept: A surface wave can be described by its height (or amplitude), which can be measured from the free-surface elevation $\eta(t)$. By analysing wave signals at different locations, we can distinguish between:

- **Incident wave height (H_i):** the height of the incoming wave before it interacts with the OWC,
- **Reflected wave height (H_r):** the portion of the wave that is reflected back upstream by the structure,

- **Transmitted wave height (H_t):** the portion of the wave that continues downstream after interacting with the OWC.

These quantities are used to define the transmission and reflection coefficients, which describe how the wave energy is distributed.

Weeks 1–3

- Place wave probes at selected locations (at least two upstream and one downstream of the chamber).
- Record free-surface elevation $\eta(t)$ at these points.
- Develop scripts (MATLAB or Python) to process the time-series data.
- Estimate wave heights (H_i , H_r , H_t) from the recorded signals using peak detection or Fourier analysis.

Weeks 6–8

- Compute transmission and reflection coefficients using:

$$K_T = \frac{H_t}{H_i}, \quad K_R = \frac{H_r}{H_i}$$

- Perform frequency analysis (FFT) to confirm the dominant wave frequency.
- Estimate wave energy flux where appropriate.

Deliverable

- Processed wave data and plots of $\eta(t)$.
- A table summarising incident, reflected, and transmitted wave heights for each case.

4. Integration and Final Phase

Week 5

- Combine all modules into a single working simulation.
- Ensure that all outputs follow a consistent format (e.g. CSV files with time, pressure, flow rate, and probe data).

Weeks 7–9

- Perform grid convergence checks to ensure numerical accuracy.

- Verify the overall energy balance:

$$E_{\text{in}} = E_{\text{ref}} + E_{\text{trans}} + E_{\text{absorbed}}$$

- Discuss and interpret results as a group.

Week 10

- Finalise the report and prepare the presentation.